

MAZAK VQC-20/40

LinuxCNC / Mesa Retrofit — Complete Manual Set

Machine: Mazak Vertical Quality Center VQC-20/40 | **S/N:** 060231
Original control: Mazatrol M-2 (retired) | **New control:** LinuxCNC + Mesa 7i97T / 7i49 / 7i84U
Shop power: 460 VAC 3-phase (onboard transformer tapped for 460 V)
Document: MZK-LCNC-000 — Master Guide | **Revision:** A — July 2026

Volume 0 — Master Guide & Table of Contents

How This Manual Set Is Organized

This set documents the Mazak VQC-20/40 (SN 060231) **as converted** from its original Mazatrol M-2 control to LinuxCNC with Mesa Electronics FPGA hardware. It supplements — it does not replace — the original Mazak factory manuals, which remain authoritative for the base machine's mechanics, lubrication, and original electrical drawings.

Volume	Document No.	Title	Audience
0	MZK-LCNC-000	Master Guide & Table of Contents	Everyone
1	MZK-LCNC-001	Machine Overview & Safety	Everyone — read first
2	MZK-LCNC-002	Operation Manual (LinuxCNC)	Operators
3	MZK-LCNC-003	Mesa Conversion & Installation	Integrator / electrician
4	MZK-LCNC-004	Wiring Diagrams & I/O Reference	Integrator / electrician
5	MZK-LCNC-005	Maintenance & Troubleshooting	Maintenance
6	MZK-LCNC-006	Reference, Glossary & Index	Everyone

⚠ PROJECT STATUS NOTICE. As of this revision the retrofit is in the **planning / bring-up stage**. Most wiring assignments in Volume 4 carry a **PLAN_VERIFY** status: they are engineered plans that **must be traced and confirmed in the cabinet** before wires are landed or outputs energized. Statuses are printed with every pin table. Nothing in this set overrides the verification steps in Volume 3.

Master Table of Contents

Volume 1 — Machine Overview & Safety

- About the Machine — VQC-20/40 SN 060231
- The Conversion at a Glance — what changed, what stayed
- General Safety Rules
- The Safety Chain (E-stop philosophy)
- Lockout / Tagout & Electrical Safety
- Retrofit-Specific Hazards (analog runaway, resolver excitation, unverified I/O states)

Volume 2 — Operation Manual (LinuxCNC)

1. Control Station Overview
2. Power-Up Sequence
3. Homing & Referencing
4. Jogging & Manual Control
5. Spindle Operation (FR-SX, gears, orient)
6. Tool Changes & the ATC / Magazine
7. Running G-code Programs
8. Coolant, Air & Auxiliary Functions
9. Feed Hold, Single Block & Cycle Control
10. Optional WHB04B Pendant
11. Shutdown Sequence
12. Operator Alarms & First Response

Volume 3 — Mesa Conversion & Installation

1. Architecture Decision Summary (7i97T + 7i49 + 7i84U)
2. Hardware Stack & Bill of Boards
3. Control PC & Ethernet Setup (hm2_eth, static IP)
4. Mesa 7i97T — role, connectors, firmware
5. Mesa 7i49 — resolver feedback interface (Tamagawa)
6. Mesa 7i84U — remote field I/O (smart-serial)
7. LinuxCNC Configuration Files (INI / HAL guide)
8. Bring-Up Procedure (8 stages)
9. Verification Checklists (pre-power, pre-motion, pre-ATC)
10. Rejected/Historical Architecture (6i25 + 7i77)

Volume 4 — Wiring Diagrams & I/O Reference

1. System Architecture Diagram
2. 7i97T TB3 — Servo Analog Commands & Enables
3. 7i49 — Resolver Wiring (with warnings)
4. 7i97T TB5 — Limits, Homes, E-stop, SSR Outputs
5. 7i97T TB4 ↔ 7i84U — Smart-Serial Link
6. Spindle Control — Mitsubishi FR-SX
7. E-stop / Safety Chain Concept
8. 7i84U TB1 — Complete Input Map (32 inputs)
9. 7i84U TB2 — Complete Output Map (16 outputs)
10. Legend, Status Codes & Field Devices

Volume 5 — Maintenance & Troubleshooting

1. Maintenance Philosophy & Original Manual Cross-Reference
2. Daily / Weekly / Monthly Schedules
3. Lubrication & Hydraulics
4. Servo Drives & Resolver Feedback Health
5. Mesa Board & LinuxCNC Health
6. Troubleshooting — Motion
7. Troubleshooting — Spindle
8. Troubleshooting — ATC / Magazine
9. Troubleshooting — Mesa Comms / HAL
10. Spare Parts & Sourcing

Volume 6 — Reference, Glossary & Index

- 1. Original Mazak Manual Library (locations & applicability)
- 2. Key Project Files (repo & OneDrive paths)
- 3. External References
- 4. Glossary of Terms
- 5. Alphabetical Index (full set)

Source Documents

This manual set was compiled July 2026 from these authoritative project sources:

Source	Location
Original Mazak manuals (11 PDFs)	OneDrive–Personal/OneDrive Docs/Mazak/Manuals_SN060231/
Architecture decision	GitHub/Github LinuxCNC/docs/architecture_decision.md
Project status & TODO	GitHub/Github LinuxCNC/docs/project_status.md
Current pin authority (91 signals)	GitHub/Github LinuxCNC/mesa/current_pin_authority.csv
Wiring master workbook	GitHub/Github LinuxCNC/wiring/Mazak_Wiring_Master_7i97T_7i49_7i84U.xlsx
HAL / INI skeletons	GitHub/Github LinuxCNC/linuxcnc/
PLC signal dictionary (385 signals)	Manuals_SN060231/02_Maintenance/YM2V39L_element_list.csv

End of Volume 0.